Department of Civil Engineering
Academic Year 2022– 2023 (Even Semester)
Degree, Semester & Branch: IV semester B.E Civil Engineering
Course Code & Title: CE3402 & Strength of Materials
Name of the Faculty member (s): Mr.V.Ragavan

Innovative Practice Description

- **Unit / Topic: Unit III / Deflection – Simply supported Beam**
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO7
- **Activity Chosen: Theory to Practical**
- **Justification:**
  The calculation of deflection value for simply supported beam as learnt as their theory subjects. So in order to give a practical exposure students are taken to strength of materials laboratory to show the experimental setup.
- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation:**
  Theoretical and practical knowledge are interconnected and complement each other — if one knows exactly HOW to do something, one must be able to apply these skills and therefore succeed in practical knowledge.
- **CO – PO / PSO mapping:**

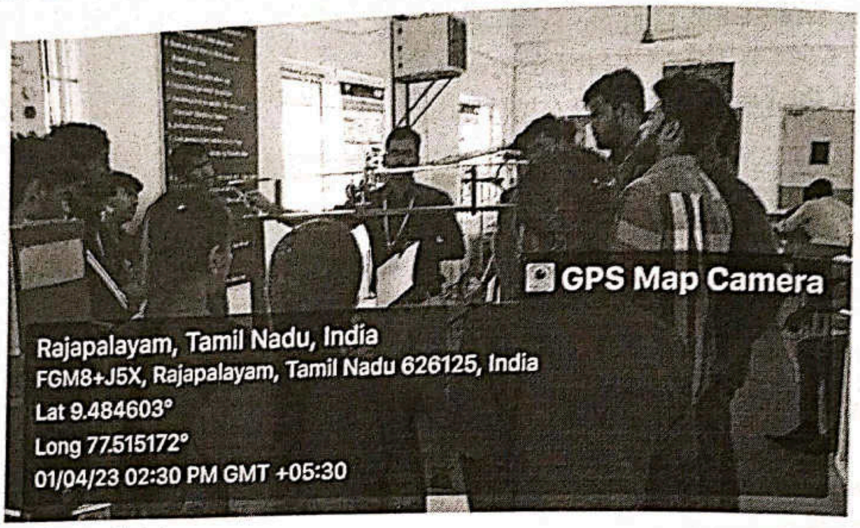
CO	PO1	PSO2
CO1	3	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PSO2
	3	3
Justification for correlation	To solve the problem the student will apply mathematical, science and engineering fundamentals	Calculate the deflection value of simply supported beam under the central point load condition.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    From this activity, the students have given the feedback as made to see lively the how to determine the deflection of beam.
  - ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - i. This activity was important to help the students to identify the bridge the gap between oral learning and hands-on experience.
    - ii. The students will be to identifying the types of beam and to calculate the deflection value of simply supported beam.
  - ❖ ***Challenges faced in implementation:***
    Initially, I have planned the activity for 15 minutes but this activity extended more than 20 minutes to determine the deflection value of simply supported steel beam.

References:

❖ Rajput R.K. "Strength of Materials (Mechanics of Solids)", S.Chand & company Ltd., New Delhi, 2018.

❖ Timoshenko.S.B. and Gere.J.M, "Mechanics of Materials", Van Nos Reinbhold, New Delhi 1999.

Signature of Faculty Member

HOD